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MILITARY SPECIFICATION

AMPLIFIER AND POWER SUPPLY,
COORDINATE CONVERTER, TYPE A-1

1. SCOPE

1.1 This specification establishes the requirements for one type of coordinate converter amplifier and power supply designated Type A-1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 The following specification, standard, and drawing, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-5935	Bombing Navigational Systems, Type K-Series, General Specification For

STANDARDS

Military

MIL-STD-129	Marking of Shipments
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

General Mills

SYA-2,000	Amplifier and Power Supply, Coordinate Converter, Type A-1

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements.- The requirements of Specification MIL-B-5935 and General Mills Drawing No. SYA-2,000 are applicable as requirements of this specification.

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3.2 Where the requirements of Specification MIL-B-5935 and this specification conflict, the requirements of this specification shall govern.

3.3 Components.- The amplifier and power supply, Type A-1, shall consist of seven plug-in amplifiers and a power supply. One complete assembly consists of the following:

7 each - Hermetically sealed plug-in Servo Amplifiers, Servomechanisms Inc., Type SA104-H, or equal.

1 each - Power Supply, Servomechanism Inc., Type PS 111, or equal.

1 each - Rack and Cable Assembly for mounting the 7 amplifiers and one Power Supply Unit.

3.4 Design.- The amplifier and power supply, Type A-1, shall be designed to operate in conjunction with the coordinate converter and periscopic bomb sight used in the applicable bombing navigational system.

3.4.1 The amplifier, transformers, and capacitors shall be hermetically sealed.

3.4.2 Mounting.- The amplifier and power supply, Type A-1 shall be designed to be mounted on a rack in a vertical position for service use having the instruments, fuses and amplifiers accessible from the front side. The cable connection shall be taken from a connector located on the top surface of the rack.

3.4.2.1 The mounting rack shall be equipped with suitable means to cushion the amplifier and power supply unit against vibration or shock in all directions.

3.4.3 Equipment.- A meter, fuses, connector and indicator shall be furnished as required.

3.5 Performance.- The amplifier and power supply, Type A-1, shall operate with performance characteristics required by the coordinate converter and the horizontal periscopic bomb sight used in the applicable bombing navigational system.

3.5.1 Individual amplifier components shall have the following electrical performance characteristics.

3.5.1.1 Power Requirements:

250 Volts DC	150 Milliamperes Max.	27.5 Watts
200 Volts DC	2 Milliamperes Max.	0.4 Watts
6.3 Volts AC	1.45 Amperes Nominal	9.1 Watts
	Total Power Input	47.0 Watts

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3.5.1.2 Input Characteristics:

Input impedance - 400,000 ohms
Signal Frequency - 400 cycles per second ± 5 percent
Saturation Signal - Approximately 300 millivolts

3.5.1.3 Output Characteristics:

Load power factor - 0.5
Load voltage, Maximum - 115 Volts
Load Current, Maximum - 200 Milliamperes
Power Output Maximum - 9 Watts

3.5.2 The power supply components shall have the following electrical characteristics:

3.5.2.1 Input Requirements:

Unregulated Source
Voltage - 115 Volts ± 5 Volts
Frequency - 380 - 980 cycles per second
Volt-ampere - 400 volt-amperes

Regulated Source
Voltage - 115 Volts ± 5 Volts
Frequency - 400 cycles ± 5 percent
Volt-amperes - 115 volt-amperes

D. C. Source
Voltage = + 27.5 D.C. ± 1.5 Volts
Current = .6 Amperes
Watts = 16.5 Watts

3.5.2.2 Output Characteristics:

3.5.2.2.1 + 250 Volt D.C. Output:

Voltage - 210 Volts - 290 Volts
Current - 350 Milliamperes Maximum continuous
1.0 Ampere Maximum continuous
Ripple - 1.0 Volt Maximum at 350 Milliamperes load.

3.5.2.2.2 + 200 Volt D.C. Output (seven outputs):

Voltage - 160 Volts - 240 Volts
Current - 14 Milliamperes total Maximum, 2 Milliamperes each output
Ripple - 300 Millivolts Maximum at 1.0 Ampere load of + 250 Volt output

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3.5.2.2.3 Heater Voltage Output:

Voltage - 6.3 ac nominal
Frequency - 400 cycles per second ± 5 percent
Current - 10 amperes

3.5.2.3 Meter Indication:

3.5.2.3.1 Green Sector:

Upper limit - 300 Volts
Lower limit - 200 Volts

3.6 Dimensions.- The components shall conform to the dimensions and configurations shown on General Mills Drawing No. SYA-2,000.

3.7 Weight.- The weight of the amplifier and power supply assembly shall not exceed 60 pounds.

3.8 Identification of Product.- Equipment, assemblies and parts shall be marked for identification in accordance with Standard MIL-STD-130. The identification date applied to the amplifier shall be as follows:

Amplifier and Power Supply, Coordinate Converter
Type A-1
Specification MIL-A-5934
Mfr's Part No. *
Serial No. *
Order No. *
Stock No. *
(Mfr's Name or Trademark)
U. S. Property

* Applicable date to be entered by the contractor.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 The inspection and testing of the amplifier and power supply assembly shall be in accordance with Specification MIL-B-5935 and as specified herein.

4.2 Preproduction Tests:

4.2.1 Preproduction Test Sample.- One amplifier and power supply assembly shall be subjected to the preproduction tests specified in Specification MIL-B-5935.

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4.3 Inspection Tests:

4.3.1 Individual Tests.- Each amplifier and power supply assembly shall be subjected to the individual tests specified in Specification MIL-B-5935 and the following:

4.3.1.1 Bench Functional Tests:

4.3.1.1.1 The amplifier and power supply shall have a complete continuity test performed on it to determine that the internal wiring, component interconnection, and external leads are correct.

4.3.1.1.2 The amplifier component shall undergo static performance tests to check against performance as approved by the procuring agency.

4.3.1.1.3 The power supply component shall undergo load tests to check against performance as stipulated herein.

4.3.1.1.4 The amplifier and power supply shall be interconnected with a complete applicable bombing navigational system or a suitable simulator and a sufficient number of simulated problems shall be run to insure that the amplifier and power supply are functioning correctly with respect to the over-all system or simulator.

4.3.2 Sampling Tests.- Selection of the units for the sampling tests shall be the first production unit of the order or the first unit following a production gap of 31 or more days; random selection of 1 unit from the next 5 units on the order; random selection from the next 10 units on the order; random selection of 1 unit from the next 15 units on the orders, and 1 unit selected at random from each lot of 30 thereafter, or 1 month's production, whichever is greater. For subsequent orders where continuous production is maintained, sampling tests shall be conducted on the basis of 1 unit selected at random from each lot of 30, or 1 month's production, whichever is greater.

4.3.2.1 When a unit that is selected as representative of the lot fails to meet the requirements, immediate action to determine the case of failure shall be taken. In addition, major functional failures of any unit during sampling tests shall require that the sampling plan outlined in 4.3.2 will be repeated after correction of the failure. This sampling plan, however, need only be repeated for the particular sampling test producing the failure and such other tests as are affected.

4.3.2.2 The sampling test shall consist of the sampling tests specified in MIL-B-5935.

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5. PREPARATION FOR DELIVERY

5.1 General.- The requirements specified herein apply only to direct purchases by or direct shipment to the Government.

5.2 Preservation, packaging, and packing requirements shall be as specified in Specification MIL-B-5935.

5.3 Marking of Shipments.- Interior packages and exterior shipping containers shall be marked for shipment in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Amplifier and Power Supply, Coordinate Converter, Type A-1 (* Mfr's Part No.), Specification MIL-A-5934.

* Applicable data to be entered by the contractor.

6. NOTES

6.1 Use.- The Type A-1 Amplifier and Power Supply covered by this specification is one component of a periscopic bombsight group and is used to amplify servo signals for operation of the applicable horizontal periscopic bombsight and coordinate converter.

6.2 Ordering Data.- Invitation for bids, contracts, and purchase orders should state the conditions for the following:

6.2.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that one amplifier and power supply will be required as a preproduction sample and that this preproduction sample will be subjected to the Preproduction Tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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